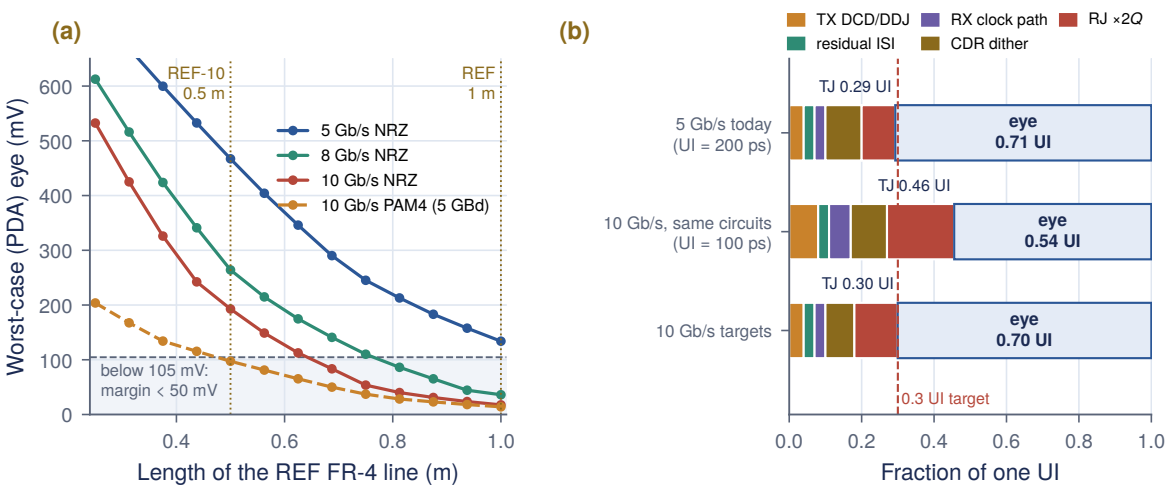
(a)
REF-10
0.5 m
REF
1 m
5 Gb/s NRZ
8 Gb/s NRZ
10 Gb/s NRZ
10 Gb/s PAM4 (5 GBd)
below 105 mV:
margin < 50 mV
Worst-case (PDA) eye (mV)
600
500
400
300
200
100
0
0.4
0.6
0.8
1.0
Length of the REF FR-4 line (m)
(b)
TX DCD/DDJ
RX clock path
RJ ×2Q
residual ISI
CDR dither
TJ 0.29 UI
5 Gb/s today
(UI = 200 ps)
eye
0.71 UI
TJ 0.46 UI
10 Gb/s, same circuits
(UI = 100 ps)
eye
0.54 UI
TJ 0.30 UI
10 Gb/s targets
eye
0.70 UI
0.3 UI target
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of one UI